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Value of Nurse Practitioner Inpatient Hospital Staffing

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Abstract

Background: Evidence indicates hospitals with better RN staffing have better patient outcomes. Whether involving more nurse practitioners (NPs) in inpatient care produces better outcomes is largely unknown.

Objective: To determine whether the presence of more NPs produces better inpatient outcomes net of RN staffing.

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Research Design: 2015-2016 cross-sectional data on 579 hospitals linked from: 1) RN4CAST-US nurse surveys; 2) HCAHPS patient surveys; 3) surgical patient discharge abstracts; 4) Medicare Spending Per Beneficiary (MSPB) reports; 5) American Hospital Association Annual Survey. Hospitals were grouped according to their NP/beds ratios (<1NP/100 beds (N=132), 1-2.99 NPs/100 beds (N=279), and 3+ NPs/100 beds (N=168).

Subjects: Registered nurses randomly sampled nurses from licensure lists yielded 22,273 RNs in study hospitals; discharge data for 1.4 million surgical patients; HCAHPS data for 86% of study hospitals.

Measures: Mortality, readmissions, lengths of stay, Medicare spending (MSPB), patient experience and quality reported by patients and nurses.

Results: After adjustments, patients in hospitals with 3+NPs/100 beds had lower odds than patients in hospitals with <1NP/100 beds of 30-day mortality (OR^2=0.76; 95%CI=0.67-0.82; P<.001) and 7-day readmissions (OR^2=0.90; 95%CI=0.86-0.96; P<.001), shorter average LOS (IRR^2=0.92; 95%CI=0.88-0.96; P<.001) and 5.4% lower average MSPB (CI=3.8%-7.1%). Patients and nurses in the hospitals with higher NP/bed ratios were significantly more likely to report better care quality and safety, and nurses reported lower burnout, higher job satisfaction, greater intentions of staying in their jobs.

Conclusions: Having more NPs in hospitals has favorable effects on patients, staff nurse satisfaction, and efficiency. NPs add value to existing labor resources.

Keywords

nurse practitioners; hospital quality; workforce

Physician participation in inpatient care is changing. Physicians are less likely than before to combine office-based practice with caring for their patients in the hospital. Physician coverage for inpatients has also been impacted by Medicare policy limiting growth in physician residency positions and limitations in resident work hours by the Accreditation Council for Graduate Medical Education. At the same time value-based purchasing payment policies incentivize hospitals to reduce readmissions and length of stay and improve patient experiences, all outcomes requiring health professionals to better manage clinical care outcomes. Hospitals are employing increasing numbers of nurse practitioners (NPs) to care for inpatients. [1]

Introduction

A robust research literature shows that better hospital registered nurse (RN) staffing is associated with better inpatient outcomes including lower mortality, fewer readmissions, and higher patient satisfaction.[2–7] There is also compelling evidence that patients in hospitals with higher proportions of baccalaureate-prepared RNs are more likely to experience favorable outcomes.[3,4,8,9,10] While many studies show that ambulatory patients cared for by NPs have good clinical outcomes and are satisfied with their care,[11,12] little is known about the value-added of NPs in inpatient care, especially in the context of increased employment

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of better educated RNs and some improvement in patient-to-RN staffing ratios over the past decade.[2]

A few studies, mostly limited to single teaching hospitals and/or intensive care, suggest that adding more NPs to inpatient settings may improve clinical outcomes.[13–17] Published research has yet to provide empirical results identifying specific interventions by NPs that might account for better clinical outcomes. While the roles of inpatient NPs vary by hospital and by specialty within hospitals, there are core components to NP practice likely to influence care outcomes. Physicians delegate more authority to NPs than to RNs to make timely decisions about patient care. NPs conduct more frequent medical rounds increasing surveillance capacity for early identification of potential complications. NPs are often more accessible than physicians, especially on surgical services, which should result in higher patient and nurse satisfaction. Yet, there is no research from a large representative sample of hospitals indicating whether employment of more NPs in general acute hospitals is associated with improved patient outcomes including lower mortality, better patient experience, and reduced readmissions and lengths of stay. Also, in line with the National Academy of Medicine's concern about the well-being and resilience of front-line caregivers,[18] it is not known whether more inpatient NPs decrease burnout and job dissatisfaction among bedside-care RNs.

This study's objective is to determine for the first time in a nationally representative sample of hospitals and patients whether increased NP staffing in hospitals adds value after taking into account investments hospitals have already made in RN staffing and preferential hiring of RNs with bachelor's education (BSNs).

Methods

Data are from: 1) the 2015-16 RN4CAST-US survey from 22,273 RNs in 579 hospitals ($\bar{x}$ **= 38** RNs/hospital) in four large states (California, Florida, New Jersey, and Pennsylvania), which represent over 20% of all acute care hospitals nationally and discharge 90% of all patients in the four states and over 25% of the patients in the U.S.;[2] 2) 2015-16 state patient discharge data for over 1.4 million surgical patients from these same hospitals (from the Office of Statewide Healthcare Planning and Development in CA, the Agency for Health Care Administration in FL, the Department of Health and Senior Services in NJ, and the Health Care Cost Containment Council in PA); 3) 2015 Hospital Consumer Assessment of Healthcare Providers and Systems (HCAHPS) survey of over 660,000 patients in 500 (86%) of these hospitals;[19] 4) Centers for Medicare and Medicaid Services Hospital Compare measure of Medicare Spending Per Beneficiary (MSPB);[20] and 5) the 2015 American Hospital Association (AHA) Annual Survey.

Hospitals in the four states represent over 20% of all acute care hospitals nationally and discharge 90% of all patients in the four states and over 25% of the patients in the U.S. RNs in these hospitals ($\overline{X}$ = 38 RNs/hospital) are similar in characteristics to hospital nurses nationally.[2]

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Nurse survey data were obtained by mailing questionnaires to homes of RNs randomly sampled from state licensure lists. The response rate of hospital nurses is unknown because our sampling frame included all RNs without information on whether or where they were employed. In their survey responses, however, nurses told us whether and where they were employed, and hospital nurses provided their employing hospital name, enabling us to link respondents to hospitals yielding a comprehensive database of almost all hospitals with 100+ beds in the four states. In an intensive re-survey of non-respondents that used extensive re-contacts, monetary incentives, and a shorter version of the original questionnaire, we were able to obtain a final response rate of 87% of the sample frame, and hence have "nearly unbiased"[21] estimates of population quantities.[22] Comparisons of nurses' responses in the main survey with those of resurveyed nonrespondents revealed no statistically significant differences at the nurse-level, suggesting no evidence of response bias.[22]

The nurse surveys asked nurses to estimate the number of NPs working in their hospital. We took the median number of NPs reported by all RNs in each hospital, divided it by the number of beds in each hospital, and grouped hospitals into three categories: <1 NP/100 beds (N=132), 1-2.99 NPs/100 beds (N=279), and 3+ NPs/100 beds (N=168). While these categories are somewhat crude and arbitrary, sensitivity analyses showed that they produced similar results to those obtained using an expanded number of categories or even a continuous measure of NPs/100 beds, the latter of which we felt would convey a false degree of precision.

Nurse survey data also provided outcome measures and additional hospital characteristics used as controls. Nurse-reported outcomes included their rating of and willingness to recommend their hospital, reports of care quality and patient safety, and their confidence that management will act to resolve patient care problems and that patients can manage their care after discharge. Nurse outcomes included whether nurses scored high on burnout (using the Maslach emotional exhaustion scale),[23] their job satisfaction and intention to remain in their job for the next year. HCAHPS patient surveys provided information on care quality in these same hospitals from the patient's perspective.[19] Medicare Spending Per Beneficiary (MSPB), an indicator of hospital efficiency, was measured as price-standardized and risk-adjusted Medicare payments for all services provided to beneficiaries spanning 3 days prior to hospital admission to 30 days following discharge.[20] At the hospital level, this measure is expressed as a ratio of the hospital's median spending per Medicare beneficiary divided by the median MSPB for all hospitals nationally, weighted by the number of hospital stays. A value of 1 suggests the hospital's MSPB equals the national median, a value greater than 1 suggests MSPB is greater than the median, and less than 1 suggests MSPB is lower than the median.

We used nurse-level data to first show percentages of nurses reporting different outcomes overall, and to test (using chi-square statistics) for differences across categories of hospitals with different NP/100 bed ratios. Then, after converting the percentage (P) of nurses reporting different outcomes to odds, which are simply (P)÷(100-P), we used robust logistic regression models to account for clustering of nurses within hospitals and estimate how the odds of nurses reporting outcomes were affected by having more NPs per 100 beds, after adjusting for other hospital and nurse characteristics. The adjusted models take account of

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RN's age, sex, and type of unit they worked on, and hospital's RN staffing, percentage of BSN nurses, number of beds, teaching status, technology, and location (state). To simplify our results, we coded the three NP/100 Beds categories 1, 2, and 3 and constrained the differences between categories to be linear. This linear constraint implies that the difference in the odds of the various outcomes is the same, and can be described by the same odds ratio (OR), when "adjacent" NP/beds categories are compared (or the same when we compare the third NP/100 beds category with the second NP/100 beds category as when we compare the second NP/100 beds category with the first NP/100 beds category. It also implies that the difference between the third and first NP/100 beds categories equals OR x OR, or OR^2.

We used hospital-level HCAHPS data, which are risk-adjusted before being publicly reported. The final HCAHPS patient-mix adjustment (PMA) model includes eight variables: self-reported health status, education, service line (medical, surgical, or maternity care), age, Emergency Room (ER) admission source, relative lag time of response, service line by age interactions, and primary language other than English. We first show the average (mean) percentages of patients reporting different outcomes across all hospitals, and then across categories of hospitals with different NP/100 Bed ratios, and tested the significance of the differences. Linear regression models were used to estimate how much the average percentage of patients reporting various outcomes differed as a result of having more NPs per 100 beds, before and after adjusting for other hospital characteristics. Here too we constrained the effect of the NP/beds ratio to be linear to simplify results, since there were no significant departures from linearity detected. We used the same series of linear regression models to estimate unadjusted and adjusted effects of categories of hospitals with different NP/100 bed ratios on hospital MSPB.

To evaluate mortality and readmissions, we used robust logistic regression models to estimate the effect of NP/100 Bed ratios on the odds of patients dying within 30 days of admission, and the odds of being readmitted to the same hospital within 7 days after discharge. To evaluate LOS, which is a count variable as opposed to a binary variable, we used zero-truncated negative binomial regression models to estimate the effect of NP/100 Bed ratios on the average LOS, which are expressed as incident rate ratios (IRRs) rather than odds ratios. The zero-truncated negative binomial regression models were necessary because the LOS measure included no zeroes (all patients admitted to the hospital were coded as having had a stay of at least one day) and was overdispersed (i.e., the variance in LOS across patients was greater than the mean LOS). Both types of models were estimated taking account of the clustering of patients in hospitals. We restricted our analyses to surgical patients' first admissions within the study period since there are well established risk-adjustment methods for them,[24–26] and treated the NP/Beds ratio as we do in the other analyses, coding the three categories 1, 2, and 3 and constraining differences between them to be linear. We show odds ratios (for mortality and re-admissions), incident rate ratios (for LOS) and the associated p-values and 95% confidence intervals for the unadjusted NP/beds effect on the three outcomes from simple bivariate models, and then re-estimate them using adjusted models which control for patient and hospital characteristics. Adjusted models included the same hospital characteristics included in models for nurse and patient outcomes. Patient characteristics included age, sex, transfer status, and dummy variables

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for 27 comorbidities from the Elixhauser Comorbidity Index and for 57 diagnosis-related groups (DRGs) for surgical patients.

Results

Table 1 shows nurse-reported outcomes (top panel), patient-reported outcomes (middle panel), and MSPB (bottom panel) across all hospitals and in the three groups of hospitals with different NP/Bed ratios. When all hospitals are considered together, 32% of nurses rate their hospital highly, 45% would definitely recommend it to others, 40% rate quality of care as “excellent,” and 46% are confident that management will act to resolve patient care problems (46%). Half the nurses (51%) are confident patients can manage their care after discharge, and 70% and 71% grade their hospitals as “A” or “B” on patient safety and infection prevention, respectively. 31% of the nurses scored in the high burnout range, 79% were satisfied with their job, and 87% intended to stay for the next year. More patients than nurses rate their hospital highly (68%) and would definitely recommend it (70%), and the majority offer favorable reports on all but one of the outcomes.

Favorable reports for all outcomes were more likely from nurses and patients in hospitals with more NPs/100 beds than in hospitals with fewer. Unadjusted differences across the three groups of hospitals are not always monotonic, but in most cases they are nearly so, and they are significant ($p<.05$) for virtually all outcomes. 37% of nurses in hospitals with 3+ NP/100 beds rate their hospital highly, while only 27% of nurses in hospitals with <1 NP/100 beds do so. While 51% of nurses in hospitals with 3+ NP/100 beds would definitely recommend their hospital highly, the same is true for only 38% of nurses in hospitals with <1 NP/100 beds. Similar but somewhat smaller differences are found for patients.

Medicare spending was significantly lower relative to the national median for hospitals with more NPs/100 beds. The mean MSPB for hospitals with 3+ NP/100 beds was 0.993 compared to 1.015 for hospitals with 1-2.9 NP/100 beds, and 1.019 for hospitals with <1 NP/100 beds.

The top panel of Table 2 shows odds ratios which estimate how the odds differ linearly across the three NP/beds categories, first estimating them using bivariate logistic regression models and then using multivariate models to adjust for other hospital and nurse characteristics. The odds ratios (ORs) estimating these differences are in all cases larger after adjustment than before, and virtually all adjusted ORs are statistically significant and sizable. The adjusted ORs indicate the difference between adjacent NP/beds categories. Taken to the 2nd power, as in the last column of the table, they indicate the differences between nurses in hospitals with 3+NP/100 beds and hospitals with <1NP/100 beds. These “squared” odds ratios indicate differences in the odds of favorable reports in the highest and lowest NP/bed categories ranging from roughly 1.3 to 2.1 for 9 of the 10 outcomes. High burnout is lower for the highest than for the lowest NP/beds category, by a factor of 0.85.

The middle panel of Table 2 shows the results for the patient-reported outcomes. Here too we constrain the differences to be linear, though here estimates are from robust regression models rather that logistic regression models, since HCAHPS data are reported for patents

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at the hospital level. These coefficients are additive and reveal how much the average percentage of patients providing favorable responses differs between adjacent categories of hospitals defined by NP/bed ratios, before and after adjusting for other hospital differences. All differences are significant except for the item related to whether staff always explained medications. Multiplied by two, as in the last column of the table, they indicate that the size of the average significant differences in those percentages between hospitals with 3+NPs/100 beds and hospitals with <1NP/100 beds range from roughly two to six percentage points.

The bottom panel of Table 2 shows results from models estimating the relationship between NP/100 bed ratios and hospital MSPB. The coefficients are additive and again represent linear differences, like those used to estimate the patient-reported outcomes models. The coefficients reflect the difference in the MSPB between adjacent groups of hospitals based on NP/100 bed ratios. Results from both unadjusted and adjusted are statistically significant and suggest a lower MSPB for hospitals with more NPs/100 beds. Again, multiplying the coefficient by two reveals the difference in MSPB (−0.054), for hospitals with 3+NPs/100 beds compared to hospitals with <1NP/100 beds. Thus, hospitals with 3+ NPs/100 beds spend on average roughly 5% less per Medicare beneficiary than hospitals with <1NP/100 beds.

Table 3 shows the linear effect of the 3-category NP/100 beds variable on 30-day mortality, 7-day readmission, and average LOS before and after adjusting for patient characteristics and for RN staffing, percentage of BSN nurses, and other hospital characteristics. For mortality and readmissions, the odds ratios for the fully adjusted Model 3 indicate that, net of the effects of the roughly 100 variables we controlled for, patients in hospitals with higher NP/beds ratios have lower odds on dying and of readmission, by factors of 0.89 and 0.95, respectively. Taken to the 2nd power, they indicate that patients in hospitals with 3+NP/100 beds have lower odds of dying and being re-admitted than patients in hospitals with <1NP/100 beds, by factors of 0.79 and 0.90, respectively. The adjusted incident rate ratio (IRR) of 0.96 implies that patients in hospitals with 1-2.9 NPs/100 beds would have lengths of stay that were, on average, lower than patients in hospitals with <1 NP/100 beds, and that patients in hospitals with 3+ NPs/100 beds would have lengths of stay that were, on average, lower than patients in hospitals with 1-2.9 NPs/100 beds, in both cases by factors of 0.96, or by 4%. Patients in hospitals with 3+ NPs/100 beds would have lengths of stay that were, on average, lower than patients in hospitals with <1 NP/100 beds, by a factor of $0.96^2 = 0.92$, or by 8%.

The models reported included controls for teaching status which is a measure of the number of resident physicians. Additionally, we controlled for (not shown) the number of hospitalists and intensivists as reported by the American Hospital Association. This variable included as a control was not significantly associated with mortality or readmissions but was associated with length of stay, However adding it to our models does not change the size or significance of the NP effect at all for any of the outcomes.

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Discussion

This is the first study to document the value-added of inpatient NP employment in a large representative sample of hospitals. Our study of 579 hospitals shows that hospitals employing a larger number of NPs had significantly better outcomes including lower mortality, fewer readmissions, shorter lengths of stay, higher patient satisfaction, and lower Medicare spending per beneficiary in addition to favorable outcomes on other quality indicators. Better outcomes associated with greater NP employment were found even after taking into account other factors known to be associated with good outcomes including favorable patient to nurse ratios, percentage of RNs with bachelor's educational qualifications, teaching status, and high technology availability. RNs in hospitals with more NPs had significantly lower job-related burnout, higher job satisfaction, and are more likely to say they intend to remain in their jobs, all outcomes associated with better nurse retention and reductions in high cost of nurse turnover,[18,27] and offer the potential of greater stability in the supply of bedside care nurses over the coming decade when large numbers of experienced nurses are expected to retire.[28,29] These better outcomes were achieved with lower healthcare spending per episode of care. Together these findings begin to build a business case for employing NPs in inpatient care.

In the context of the study's strengths and limitations, our data are cross-sectional which limits our warrant for establishing causality, although we have previously documented that findings from cross-sectional research on the association of hospital nurse resources with patient outcomes are essentially the same as those found in a longitudinal panel study of hospitals.[2] Future panel studies would be helpful in evaluating whether increases in NP employment are associated with better outcomes. We cannot eliminate the possibility that the results we find are due to unmeasured variables but our controls are extensive and take into account characteristics of hospitals and patients that have been found in prior research to be associated with patient outcomes. We cannot decipher whether the better outcomes we find associated with higher NP employment are due to specific types of improvements in care. Our measure of NP staffing is crude (NPs per 100 beds) but is the best measure available for a large number of hospitals. If anything, our measure likely underestimates the association between hospital NP staffing and patient outcomes. In looking at objective patient outcomes (i.e., mortality, readmissions and lengths of stay), the patients we study are a group of general, orthopedic, and vascular surgery patients represented in most all acute general hospitals, most of whom do not experience an ICU stay. We have used a wider array of outcomes to evaluate the added value of inpatient NP care than any previous research including reports from tens of thousands of bedside care RNs practicing in the study hospitals, reports of patients receiving care in them, and objective clinical outcomes data on patients of the study hospitals from independent sources. Finally, in a framework of value-based purchasing, we find that hospitals that employ more nurse practitioners per 100 beds show lower healthcare spending per episode of care that extends to 30 days following discharge.

Our findings are relevant to policy debates taking place in many states regarding modernizing state practice acts governing nurse practitioner scope of practice. Twenty-two states still require NPs to have formal collaborative agreements with physicians

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including inpatient NPs who provide care in physician-rich environments as members of interdisciplinary teams. All four states studied are collaborative practice agreement states.[30] There is no evidence documenting that state-mandated collaborative agreements result in improved patient outcomes. Many NPs pay physicians to meet this regulatory requirement which adds to the cost of care. While hospitals are unlikely to pay their physicians extra to participate in state-mandated collaborative agreements, hospitals still bear substantial annual staff costs to meet state documentation requirements. Hospitals have been relatively silent on NP scope of practice policy debates but as hospitals employ more NPs, this may change.

Conclusion

The complexity of hospitalized patients continues to grow as advances in science and technology create new opportunities to extend lives and improve daily functioning for an aging population with multiple health challenges. The imperative to provide value in the expensive context of inpatient care is increasingly important. Our study shows that nurse practitioners in advanced care roles in inpatient care are a valuable adjunct to existing inpatient hospital physician and RN professional care.

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Table 1.

Nurse-Reported Outcomes across All Hospitals, and in Hospitals with Varying NP/Bed Ratios

		Hospitals in Which the NP/100 Bed Ratio is -		
Nurse-Reported Outcomes	**All Hospitals (N = 579)**	**<1 NP per 100 Beds (N= 132)**	**1-2.9 NPs per 100 Beds (N= 279)**	**3+ NPs per 100 Beds (N= 168)**
Rate hospital highly (9 or 10 on a 10-point scale)	32.0%	27.0%	30.1%	37.2%
Would definitely recommend their hospital	45.0%	38.2%	42.9%	51.3%
Report "Excellent" Quality of Care	39.5%	33.6%	37.0%	45.7%
Grade hospital "A or B" on Patient Safety	70.4%	66.5%	69.0%	74.3%
Grade hospital "A or B" on the Prevention of Infections	71.1%	68.4%	69.0%	75.3%
Confident Management Will Act to Resolve Patient Care Problems	45.9%	43.2%	44.1%	49.7%
Confident Patients Can Manage Their Care After Discharge	51.1%	50.2%	49.1%	54.5%
High burnout	31.0%	31.3%	32.2%	29.1%
Satisfied with Job	79.0%	78.7%	78.1%	80.4%
Intend to stay in current job	87.1%	87.0%	86.4%	88.1%
Patient-Reported Outcomes	**(N = 500)**	**(N = 112)**	**(N= 243)**	**(N= 145)**
Rate hospital highly (9 or 10 on a 10-point scale)	68.2%	66.4%	67.4%	70.9%
Would definitely recommend the hospital	69.7%	67.7%	68.8%	72.6%
Nurses always communicated well	76.3%	74.5%	76.0%	78.2%
Always received help as soon as they wanted	62.0%	59.7%	61.7%	64.1%
Pain was always well controlled	68.7%	67.9%	68.3%	69.9%
Staff always explained medications	61.1%	60.8%	60.6%	62.3%
Staff provided information about what to do during recovery at home	49.2%	47.7%	48.5%	51.4%
Hospital Efficiency	**(N = 496)**	**(N = 110)**	**(N= 243)**	**(N= 143)**
Medicare spending per beneficiary, mean (SD)	1.009 (0.072)	1.019 (0.067)	1.015 (0.068)	0.993 (0.08)

Sources: 2015-16 RN4CAST-USA Nurse Survey data and 2015 HCAHPS Patient Survey data.

Notes: Chi-square tests and F-tests indicate that all differences between the three groups of hospitals (with 2 degrees of freedom) are significant at the .01 level.

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Table 2.

Unadjusted and Adjusted Odds Ratios and Coefficients Estimating Differences in Nurse- and Patient-Reported Outcomes, and Medicare Spending across Hospitals with Varying NP/Bed Ratios

	Unadjusted		Adjusted		
Nurse Reported Outcomes	**OR**	**95% CI**	**OR**	**95% CI**	**OR[2]**
Rate hospital highly (9 or 10 on a 10-point scale)	1.29***	(1.16 – 1.44)	1.35***	(1.22 – 1.50)	1.82
Would definitely recommend their hospital	1.33***	(1.19 - 1.47)	1.44***	(1.30 – 1.61)	2.09
Report “Excellent” Quality of Care	1.32***	(1.21 – 1.44)	1.33***	(1.22 – 1.45)	1.77
Grade hospital “A or B” on Patient Safety	1.22***	(1.12 – 1.32)	1.26***	(1.15 – 1.37)	1.58
Grade hospital “A or B” on the Prevention of Infections	1.21***	(1.11 – 1.32)	1.23***	(1.12 – 1.35)	1.51
Confident Management Will Act to Resolve Patient Care Problems	1.16***	(1.07 – 1.27)	1.20***	(1.10 – 1.32)	1.45
Confident Patients Can Manage Their Care After Discharge	1.12**	(1.03 – 1.20)	1.17***	(1.07 – 1.27)	1.37
High burnout	0.94*	(0.88 – 0.99)	0.92*	(0.86 – 0.99)	0.85
Satisfied with Job	1.07	(0.98 – 1.15)	1.17***	(1.08 – 1.27)	1.38
Intend to stay in current job	1.07	(0.99 – 1.16)	1.14**	(1.05 – 1.25)	1.31
Patient-Reported Outcomes	**β**	**95% CI**	**β**	**95% CI**	**2 x β**
Rate hospital highly (9 or 10 on a 10-point scale)	2.33***	(1.43 – 3.23)	2.59***	(1.65 – 3.52)	5.17
Would definitely recommend the hospital	2.55***	(1.51 −3.60)	3.13***	(2.07 – 4.19)	6.25
Nurses always communicated well	1.84***	(1.34 – 2.34)	1.12***	(0.60 – 1.64)	2.24
Always received help as soon as they wanted	2.19***	(1.47 – 2.91)	1.30***	(0.53 – 2.07)	2.60
Pain was always well controlled	1.00***	(0.54 – 1.47)	0.86**	(0.35 – 1.37)	1.72
Staff always explained medications	0.79**	(0.26 – 1.32)	0.44	(−0.16 – 1.03)	0.87
Staff provided information about what to do during recovery at home	1.94***	(1.26 – 2.62)	1.84***	(1.10 – 2.58)	3.68
Hospital Efficiency	**β**	**95% CI**	**β**	**95% CI**	**2 x β**
Medicare spending per beneficiary	−0.014**	(−0.002 – −0.005)	−0.027***	(−0.353 – −0.019)	−0.054

Sources: Data are from the 2015-16 RN4CAST-USA Nurse Survey and the 2015 HCAHPS Patient Survey.

Notes: Single, double and triple asterisks indicate coefficients that are significant at the .05, .01, and .001 levels, respectively. The coefficients for models related to the nurse-reported outcomes are odds ratios estimated using robust logistic regression models. Adjusted coefficients are from multivariate models which control for nurse characteristics (age, sex, and type of unit they worked on) and hospital characteristics (patient-to-nurse ratios, percentage of BSN nurses, number of beds, teaching status, technology, and the state in which the hospital is located). The coefficients for models related to the patient-reported outcomes are additive coefficients estimated using linear regression models which use hospital-level data that are risk-adjusted for differences across hospitals in patient characteristics. Unadjusted coefficients are from bivariate models. Adjusted coefficients are from multivariate models which control for the same hospital characteristics as in the nurse outcome models.

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Table 3.

Unadjusted and Adjusted Odds Ratios (ORs) and Incident Rate Ratios (IRRs) Indicating the Effect of NPs/100 Beds on 30-Day Mortality, 7-Day Readmission, and Length of Stay

30-Day Mortality	**OR**	**(95% C.I.)**	**OR[2] (95% C.I.)**
Model 1 – Unadjusted	0.94*	(0.89 – 0.99)	
Model 2 – Adjusted for Patient Characteristics	0.89***	(0.85 – 0.93)	
Model 3 - Adjusted for Patient & Hospital Characteristics	0.89***	(0.84 – 0.94)	0.79 (0.67 - 0.82)
7-Day Readmission	**OR**	**(95% C.I.)**	**OR[2] (95% C.I.)**
Model 1 – Unadjusted	1.00	(0.97 – 1.03)	
Model 2 – Adjusted for Patient Characteristics	0.99	(0.97 – 1.01)	
Model 3 - Adjusted for Patient & Hospital Characteristics	0.95***	(0.93 – 0.97)	0.90 (0.86 - 0.96)
Length of Stay	**IRR**	**(95% C.I.)**	**IRR[2] (95% C.I.)**
Model 1 – Unadjusted	1.00	(0.97 - 1.03)	
Model 2 – Adjusted for Patient Characteristics	0.99	(0.97 - 1.01)	
Model 3 - Adjusted for Patient & Hospital Characteristics	0.96***	(0.94 - 0.98)	0.92 (0.88 - 0.96)

Sources: 2015-16 RN4CAST-USA Nurse Survey data, 2015-16 patient discharge data from the Office of Statewide Healthcare Planning and Development in CA, the Agency for Health Care Administration in FL, the Department of Health and Senior Services in NJ, and the Pennsylvania Health Care Cost Containment Council in PA, and 2015 administrative data from the American Hospital Association (AHA) Annual Survey.

Notes: Single, double and triple asterisks indicate coefficients that are significant at the .05, .01, and .001 levels, respectively. The coefficients for models are odds ratios and incident rate ratios estimated using robust logistic regression models and zero truncated negative binomial regression models. Adjusted coefficients are from multivariate models which control for patient characteristics (age, sex, and transfer status, 30 comorbidities from the Elixhauser Comorbidity Index based on the International Classification of Diseases (ICD) diagnosis codes, and codes for 57 diagnosis-related groups (DRGs) for surgical patients) and hospital characteristics (patient-to-nurse ratios, percentage of BSN nurses, number of beds, teaching status, technology, and the state in which the hospital is located).